

MAIN INCOMER

DG INCOMER

BUS COUPLER

ELECTRICAL PANEL

Main L.T. Panel with ACB (Air Circuit Breaker)



OUTGOING
MCEB

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MCEB



Description About Main L.T. Panel with ACB

➤ **Supply, installation, testing and commissioning of Main L.T. Panel** fabricated from **CRCA sheet steel**, floor mounted, dust and vermin proof, suitable for indoor installation, complete with **ACB incomer**, busbars, protection, metering, wiring and accessories as per technical specifications and standards..

- Panel Construction
- Incomer Breaker (ACB)
- Busbar System
- Outgoing Feeders (if required)
- Metering & Instrumentation
- Internal Wiring
- Cable Termination
- Safety Features
- Standards
- Makes of Major Components
- Scope of Supply & Warranty

1. PANEL CONSTRUCTION

- Panel shall be **floor mounted, free standing, cubicle type**, modular design.
- Fabricated from **2.0 mm thick CRCA sheet** for structure and **1.6 mm sheet for doors and partitions**.
- Panel shall be **powder coated with epoxy polyester paint** after proper 7 tank treatment.
- Colour shade: **RAL 7035 / Siemens grey** or equivalent.
- Panel shall have **IP-42 / IP-54 protection** as required.
- Suitable for **415V, 3 Phase, 4 Wire, 50 Hz AC supply system**.

2. INCOMER BREAKER (ACB)

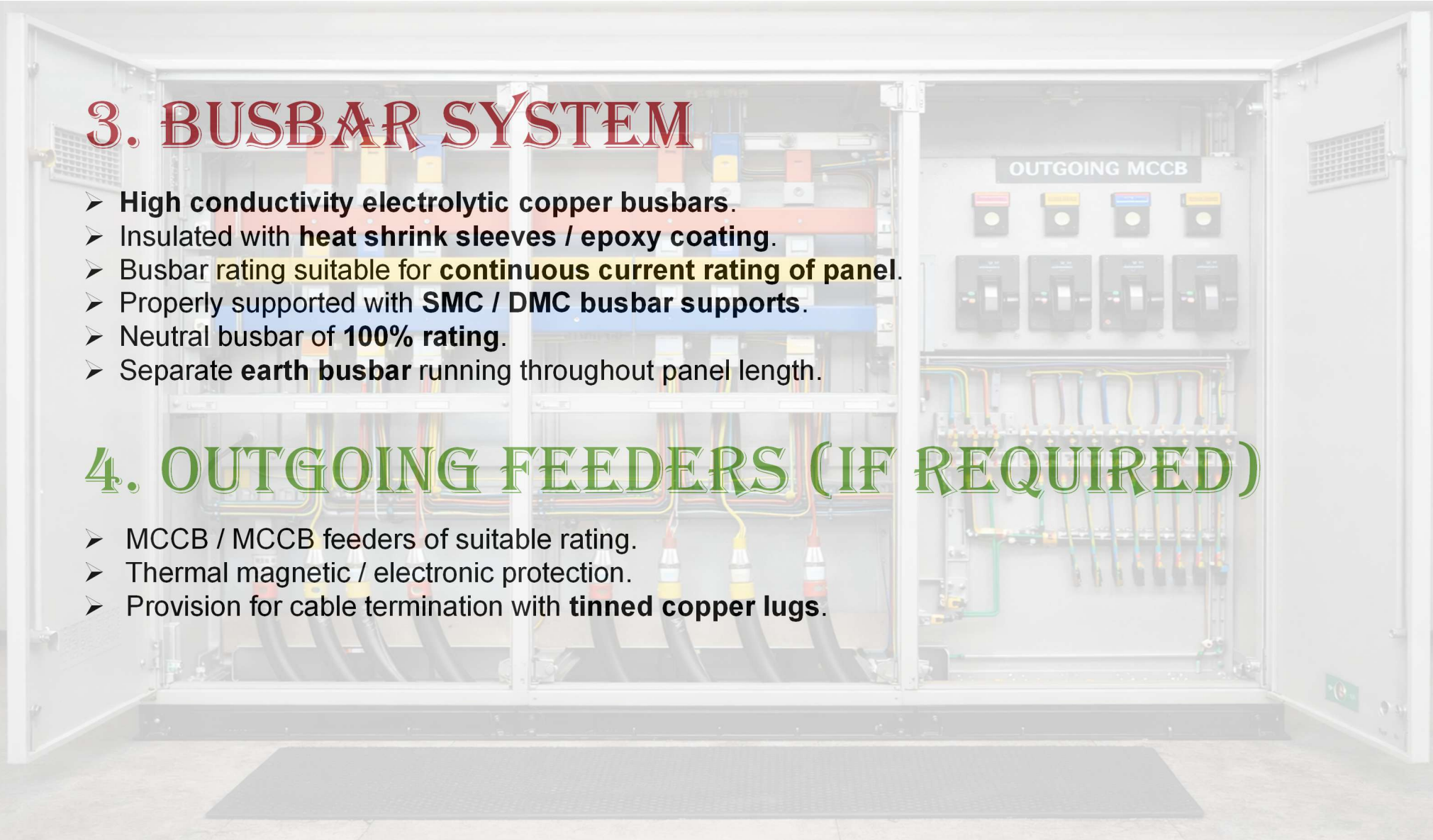
- **ACB (Air Circuit Breaker)** with following specifications:
- Rating: **800A / 1000A / 1600A** (as per requirement)
- Breaking Capacity: **50 kA / 65 kA**
- 4 Pole / 3 Pole
- Microprocessor based **LSIG protection**
- Motorised spring charging mechanism
- Electrical & mechanical trip indication
- With **shunt trip & under voltage release**
- Make:
- **Schneider Electric / Siemens / ABB / L&T**

3. BUSBAR SYSTEM

- High conductivity electrolytic copper busbars.
- Insulated with **heat shrink sleeves / epoxy coating**.
- Busbar rating suitable for **continuous current rating of panel**.
- Properly supported with **SMC / DMC busbar supports**.
- Neutral busbar of **100% rating**.
- Separate **earth busbar** running throughout panel length.

4. OUTGOING FEEDERS (IF REQUIRED)

- MCCB / MCCB feeders of suitable rating.
- Thermal magnetic / electronic protection.
- Provision for cable termination with **tinned copper lugs**.



5. METERING & INSTRUMENTATION

Panel shall include:

➤ Digital multifunction meter

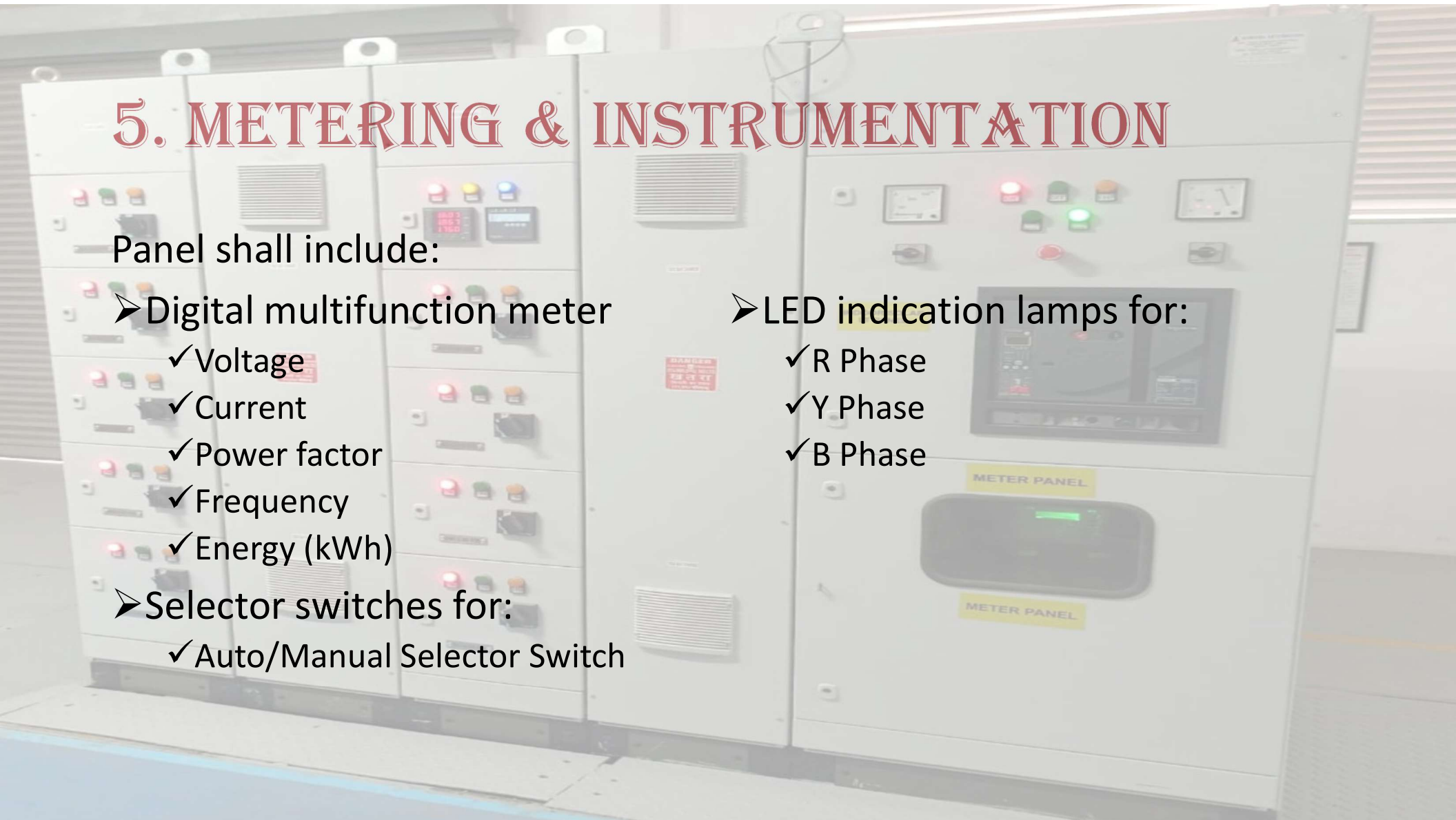
- ✓ Voltage
- ✓ Current
- ✓ Power factor
- ✓ Frequency
- ✓ Energy (kWh)

➤ Selector switches for:

- ✓ Auto/Manual Selector Switch

➤ LED indication lamps for:

- ✓ R Phase
- ✓ Y Phase
- ✓ B Phase

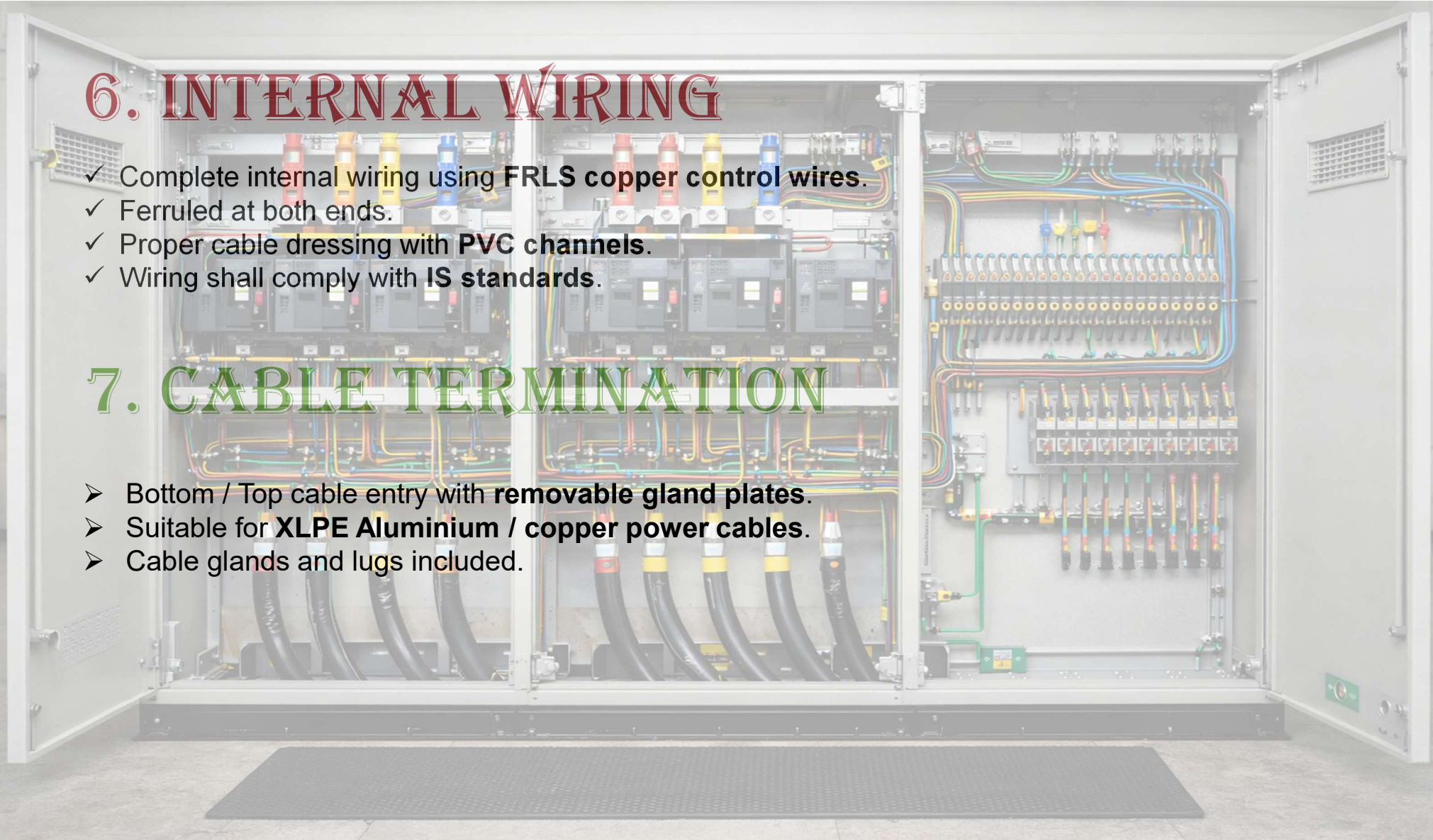


6. INTERNAL WIRING

- ✓ Complete internal wiring using **FRLS copper control wires**.
- ✓ Ferruled at both ends.
- ✓ Proper cable dressing with **PVC channels**.
- ✓ Wiring shall comply with **IS standards**.

7. CABLE TERMINATION

- Bottom / Top cable entry with **removable gland plates**.
- Suitable for **XLPE Aluminium / copper power cables**.
- Cable glands and lugs included.



8. SAFETY FEATURES

- ✓ Proper **earthing arrangement**.
- ✓ Interlocking between **ACB door and breaker**.
- ✓ Danger notice plate.
- ✓ Rubber mat in front of panel (optional).



9. STANDARDS

Panel shall conform to:

- **IS 8623 / IEC 61439**
- **IS 13947**
- Latest electrical safety standards.

SAFETY FEATURE & STANDARD

- Proper Earthing Arrangement
- Shunt Trip & Under Voltage Release
- ACB Door Interlock
- Danger Notice Plate
- Rubber Mat



MAIN LT PANEL COMPONENT MAKES & WARRANTY

10. MAKES OF MAJOR COMPONENTS

COMPONENT	APPROVED MAKE
➤ ACB	✓ Schneider / Siemens / ABB / L&T
➤ MCCB	✓ Schneider / Siemens / ABB / L&T
➤ Meter	✓ Secure / L&T / Schneider/NIPPON/MULTISPAN
➤ Indication Lamps	✓ Teknic / L&T/PECOX
➤ Switches	✓ Kaycee / L&T

11. SCOPE OF SUPPLY & WARRANTY

THE SCOPE SHALL INCLUDE :-

- Design
- Manufacturing
- Supply
- Installation
- Testing
- Commissioning
- Documentation
- Warranty.

WARRANTY :-

Minimum **12 months warranty** from date of commissioning.

